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FORM 2

THE PATENTS ACT, 1970

(39 of 1970)

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THE PATENTS RULES, 2016

PROVISIONAL SPECIFICATION

(See Section 10 and rule 13)

AN INTEGRATED MONITORING DEVICE, FOR ASSESSING AND PREDICTING THE POSSIBILITY OF OCCURRENCE OF AT LEAST FIVE MAJOR DISEASES INCLUDING BREAST CANCER, LIVER DISEASE, KIDNEY DISEASE, HEART DISEASE, AND DIABETES

Applicant(s): Pranay Sadani

Nationality: INDIAN

Address: 5B Russel Street, Park Street Area, Kolkata – 700071, West Bengal

Applicant(s): Mrityunjay Gupta

Nationality: INDIAN

Address: 8JWG+J6H, 37-38, Old RTO Rd, Gandhi Nagar, Bhilwara, Rajasthan 311001

The following specification particularly describes the nature of the invention

1. Field of the Invention

The present invention relates to computer-implemented processing of biochemical laboratory measurements and artificial-intelligence-assisted assessment of multiple disease-associated outcomes. More particularly, the invention discloses a unified blood-and-urine laboratory data processing arrangement in which five independently developed predictive models assess diabetes, kidney disease, liver disease, cardiovascular disease and breast cancer using coordinated observation validation, biochemical feature generation, fitted-model inference and integrated presentation of disease-specific results.

The invention further relates to a compatibility-controlled execution arrangement for sharing validated laboratory features and reusable numerical transformations among independently trained models while preserving the input requirements, prediction functions, calibration parameters and outcome definitions applicable to each disease.

2. Background of the Invention

Biochemical laboratory investigations provide a broad range of observations relevant to metabolic, renal, hepatic, cardiovascular and hematological conditions. Such observations are conventionally recorded through independent assays and laboratory reporting systems. Their computational interpretation may involve heterogeneous measurement units, distinct specimen types, incomplete observations, differences in collection times and multiple derived biochemical parameters.

Artificial intelligence and machine learning permit the recognition of complex associations between measured biochemical parameters and disease-related outcomes. Conventionally, individual disease-prediction models are developed and deployed through separate data-preparation, numerical-transformation and inference pipelines. When multiple models use related biochemical inputs, these independent pipelines can repeat unit conversions, input-quality verification, calculation of derived indicators and assembly of machine-readable features.

The present invention addresses these technical processing requirements through a unified observation representation, a conditional biochemical feature engine, an independently verifiable model registry and a request-scoped mechanism for reusing compatible fitted transformations. Each disease assessment is produced by an independent predictive model using a disease-specific feature schema and model-fitted preprocessing arrangement.

3. Objectives of the Invention

One objective of the invention is to provide a unified software-based system for generating five separate disease-associated assessments from existing blood and urine laboratory observations.

Another objective of the invention is to establish a common validated representation of laboratory observations through unit conversion, specimen verification, collection-time checks and measurement-quality processing.

A further objective of the invention is to calculate reusable biochemical features, including inflammatory ratios, metabolic indices, lipid measures and renal or hepatic indicators, according to defined input dependencies.

Yet another objective of the invention is to preserve independently fitted predictive models, disease-specific ordered input schemas, imputation settings, scaling parameters and calibration functions within a versioned model registry.

One more objective of the invention is to reduce repetitive numerical processing by identifying compatible fitted transformation states and reusing previously generated feature vectors within an individual processing request.

A further objective is to furnish a single integrated report containing separately labeled model scores, associated source endpoints, input-coverage information and machine-readable processing status.

4. Brief Description of the Invention

Essentially, the present invention discloses a computer-implemented integrated laboratory assessment system comprising: (a) a laboratory observation input interface configured to receive existing blood and urine measurements together with associated units, specimens, collection metadata and quality information; (b) a canonical observation processor configured to validate and transform the received measurements; (c) a biochemical feature-generation engine configured to compute derived indicators from accepted observations; (d) a versioned registry containing independent disease-prediction artifacts and their respective input and preprocessing definitions; (e) a computational reuse controller configured to identify and share compatible feature projections and fitted numerical transformations within the processing of a patient record; and (f) a reporting interface configured to present five independently generated disease assessments.

According to one embodiment, the canonical processor retains a validated snapshot of incoming laboratory observations and a source-content fingerprint. The processor verifies the correspondence between the retained snapshot and source record before performing downstream inference.

According to another embodiment, the feature-generation engine stores accepted laboratory measurements and derived biochemical features in a request-local mapping. The engine evaluates the dependencies and permissible numerical domains for every feature recipe.

According to a further embodiment, the model registry holds ordered model-input features, artifact integrity digests, fitted preprocessing states, endpoint descriptions, input requirements, population applicability, training-domain ranges and calibration information. The registry thereby enables independent model selection and verifiable execution.

According to yet another embodiment, the computational reuse controller constructs a compatibility key from a declared feature-engine version and the fitted numerical preprocessing state. Where compatible preprocessing is encountered within the same patient request, a previously computed transformed representation is supplied to the corresponding independent predictor.

5. Detailed Description of the Invention

5.1 Laboratory observation input and canonical processing

The input record comprises a patient identifier, patient age, relevant demographic and collection metadata, and a set of keyed laboratory observations. Each observation comprises a numeric value, measurement unit, specimen category, collection timestamp, availability indicator and quality designation, together with optional reference-interval and laboratory-source metadata.

The canonical processing module applies defined observation schemas, validates numeric domains and specimen compatibility, and converts accepted measurements into common units. Illustrative transformations include blood glucose in mmol/L to mg/dL by multiplication with 18.0182, serum

creatinine in µmol/L to mg/dL by division by 88.4, and albumin in g/L to g/dL by division by ten. Ordinal urine measurements and quantitatively measured urine constituents are maintained as distinct variables.

The system separately represents urea and blood urea nitrogen and converts differential white-cell measures into internally consistent concentration units. For example, a white-cell measurement supplied in cells/mm^3 is converted to 10^9/L using a factor of 1/1000. The accepted canonical observations are retained within a validated snapshot associated with the source-record fingerprint.

5.2 Laboratory quality verification and derived-feature generation

The observation processor evaluates the supplied quality designation, temporal compatibility and cross-measurement numerical consistency. A complete differential white-cell count is evaluated against its corresponding total white-cell count using a configurable consistency tolerance. Features dependent upon accepted observations are generated in a deterministic order by the biochemical feature engine.

The feature engine computes, as applicable, neutrophil-to-lymphocyte ratio (NLR), platelet-to-lymphocyte ratio (PLR), monocyte-to-lymphocyte ratio (MLR), AST-to-ALT ratio, albumin-to-globulin ratio, triglyceride-to-HDL ratio, non-HDL cholesterol, HOMA-IR, FIB-4 and estimated glomerular filtration rate (eGFR). Each recipe applies defined dependency, unit and numerical-domain conditions. Fasting-context requirements are incorporated into HOMA-IR processing; age, creatinine-standardization and equation inputs are incorporated into the eGFR procedure.

5.3 Independent predictive models and model registry

The integrated system employs five independent trained prediction functions. Each model is stored with its own disease endpoint, ordered input schema, fitted imputation and scaling parameters, feature observation indicators, calibrated output function and model-artifact digest. A registry record identifies the corresponding artifact version and fitted-preprocessing fingerprint.

Prior to inference, the system verifies the cryptographic SHA-256 digest of the selected local model artifact and checks the loaded model against the registered feature order, endpoint and fitted preprocessing definition. The independent predictive parameters and calibrators are retained for every disease-specific output.

5.4 Compatibility-controlled transformation sharing

For a given patient request, the system first constructs the common accepted-feature mapping. Each model then requests an ordered input projection according to its registered feature schema. A transformation key is constructed from the feature-processing engine version and a digest incorporating the input feature order, imputation state, scaling state, observation-indicator configuration and recorded training-domain bounds.

The processor reuses an existing transformed feature vector whenever the corresponding transformation key matches an earlier transformation in that request. The diabetes and cardiovascular embodiments share a compatible fitted preprocessing state, permitting four distinct transformed representations to serve five independently fitted prediction functions in the disclosed implementation.

In operation, the compatibility controller verifies the integrity of the locally registered model artifact, retrieves its ordered feature contract and complete fitted preprocessing signature, and selects the corresponding request-local transformation entry. The signature incorporates the numerical-engine version, ordered feature identities, fitted imputation statistics, scaling coefficients, missing-value indicator policy and empirical training bounds. An exact signature match directs a previously calculated numerical

representation to an independently fitted predictor; each distinct signature initiates its corresponding feature projection and transformation. The source-record fingerprint is checked against the canonical snapshot before feature reuse, providing a consistent relationship between validated measurements and the model-specific numerical input.

The computational arrangement maintains an individual feature and transformation cache for each patient record, together with shared read-only fitted model artifacts. This combination enables reproducible inference and controlled reuse of numerical processing across multiple predictive components.

5.5 Input applicability, prediction and integrated reporting

Disease-specific inference uses registered input contracts to evaluate essential laboratory measurements, applicable patient metadata and input quality. The model-specific preprocessing pipeline handles available optional features through the fitted observation masks and learned imputation parameters. The system also computes measurement coverage and empirical training-range indicators for inclusion in the model output.

The model inference module applies the corresponding trained predictor to the prepared feature representation and then applies its separately fitted calibration function. Each result is associated with its originating disease endpoint and artifact metadata. The integrated report provides five disease-labeled assessment entries together with the processing information applicable to each entry.

The five result entries preserve the definitions of their respective source outcomes: physician-diagnosed diabetes status in NHANES, the UCI CKD classification, the ILPD liver-patient classification, reported historical cardiovascular disease in NHANES, and diagnosed breast cancer versus healthy-control status in the routine-blood cohort. Calibration functions are fitted separately using the corresponding calibration partitions. Each numerical score is presented with its disease-specific source endpoint and calibration provenance, enabling a unified report with independently interpretable outputs.

5.6 Software interface and data flow

In one implemented embodiment, a FastAPI service receives individual laboratory records and batches of up to 128 records. A Streamlit interface provides entry of existing blood and urine laboratory measurements, displays the returned assessments and generates a structured downloadable JSON report. Each incoming record proceeds through validation, canonical conversion, biochemical feature generation, registry verification, ordered feature projection, compatible numerical transformation, independent inference and report generation.

6. Disease-Specific Predictive Embodiments

6.1 Diabetes assessment

The diabetes model employs a gradient-boosted prediction function developed using NHANES 2017–2018 adult laboratory observations. The source endpoint is physician-diagnosed diabetes recorded using DIQ010 responses. The implemented model applies HbA1c, glucose and the associated registered laboratory features, together with applicable sex metadata, to produce a calibrated source-endpoint assessment.

6.2 Kidney disease assessment

The kidney model employs a boosted classification function developed from the UCI chronic kidney disease laboratory dataset. The model processes accepted renal chemistry, hematology and available urine-

observation features. Serum creatinine and urea are registered as principal laboratory inputs, while other source-derived observations are handled through the independently fitted preprocessing state.

6.3 Liver disease assessment

The liver model employs a separately trained multilayer perceptron for the Indian Liver Patient Dataset (ILPD) binary liver-patient classification endpoint. Input features include ALT, AST, bilirubin, albumin and related laboratory measurements. The model uses independent feature processing, trained network parameters and a disease-specific calibrated score.

6.4 Cardiovascular disease assessment

The cardiovascular model employs a logistic prediction function based on NHANES laboratory and survey data. The source endpoint represents a reported history of coronary heart disease, myocardial infarction or stroke derived from MCQ160C, MCQ160E and MCQ160F. Lipid observations, including HDL and total cholesterol, and applicable demographic metadata are incorporated in the registered feature contract.

6.5 Breast cancer assessment

The breast cancer model employs a boosted prediction function developed from a source cohort of 120 pretreatment Senegalese women, comprising 80 diagnosed breast-cancer cases and 40 healthy controls. The prediction inputs comprise age, complete blood count measurements, C-reactive protein and available inflammatory ratios. The cohort is processed using a consistent differential-count quality procedure during model development and inference.

The source cohort includes 40 triple-negative breast cancer cases, 40 hormone-dependent breast cancer cases and 40 healthy comparison participants. The held-out test subset comprises 24 participant records, including 16 cases and eight controls. The model achieved an AUROC of 0.922 on these held-out source-cohort records, using routine blood-count, C-reactive protein and derived inflammatory features.

7. Mathematical and Computational Operation

Let v_j denote a source laboratory value, u_j its unit, and C_j,u_j the canonical conversion operator. The canonical observation is $c_j = C_j,u_j(v_j)$. A derived feature $z^d = f^d(\mathrm{cdeps}(d))$ is evaluated according to its declared dependencies. Accepted features form a canonical request-local mapping x.

Representative biochemical calculations are: NLR = N/L; PLR = P/L; MLR = M/L; AST/ALT = AST ÷ ALT; albumin/globulin = albumin ÷ (total protein − albumin); TG/HDL = triglycerides ÷ HDL; non-HDL cholesterol = total cholesterol − HDL; HOMA-IR = fasting glucose × insulin ÷ 405; and FIB-4 = age × AST ÷ (platelets × √ALT). Source-value units are normalized before these operations.

For independent model k, the ordered feature vector is assembled according to the registered schema I_k. Let ν_k, μ_k and σ_k denote the corresponding fitted imputation, scaling mean and scaling factor; b_k denotes observed-value indicators. The standardized transformation $T_k(x)$ is calculated from the imputed features, scaling parameters and observation indicators.

A compatibility key $K_k = (\mathrm{vengine}, H(I_k, \nu_k, \mu_k, \sigma_k, \mathrm{indicator}_k, \ell_k, h_k))$ is generated using the feature-engine version and a digest H of the complete fitted preprocessing configuration. Compatible K_k values identify reusable transformed representations. The independent raw prediction $r_k = h\theta_k(T_k(x))$ is processed through the fitted calibration function $s_k = \mathrm{sigmoid}(\gamma_k r_k + \eta_k)$.

For comparative evaluation, sensitivity and specificity are calculated from the observed source-class labels at selected validation-derived score thresholds. The independent test-cohort AUROC, Brier score and logarithmic loss are computed for each disease-specific model.

8. Experimental Evaluation and Results

The five predictive components were evaluated using their respective original training, validation, calibration and held-out test partitions. The integrated inference service executes the retained fitted models and produces independently labeled results. Table 1 summarizes the source-cohort held-out classification results recorded for the five components.

For each independently developed model, source records were partitioned into training (40%), validation (20%), calibration (20%) and test (20%) subsets using the documented fixed randomization procedure. Fitted preprocessing parameters, including imputation and scaling statistics, were derived from training data; model selection was conducted using development-validation data and sigmoid calibration was fitted on the dedicated calibration partition. Validation-selected operating thresholds were applied to test predictions. Additional liver and cardiovascular model-family comparisons employed nested training-partition cross-validation and a validation-loss criterion for retaining the preferred fitted model. Source partitions, candidate configurations and per-record predictions are preserved with the experimental artifacts.

Disease-specific model	Test n	AUROC	95% CI	Brier score	Log loss
Diabetes	1,011	0.923	0.900–0.944	0.067	0.234
Kidney	80	1.000	1.000–1.000	0.004	0.020
Liver	114	0.760	0.676–0.838	0.174	0.499
Cardiovascular	1,039	0.785	0.739–0.827	0.085	0.287
Breast cancer	24	0.922	0.804–1.000	0.122	0.401

The AUROC of 1.000 obtained for kidney disease reflects perfect discrimination within the limited UCI CKD test cohort, potentially attributable to the strong separation of renal biomarkers used in the original disease classification. Further evaluation on larger, independent and clinically diverse datasets will assess generalizability and provide a more representative estimate of predictive performance.

The kidney AUROC confidence interval of 1.000–1.000 was calculated using conditional resampling of held-out predictions. The original test set contained 80 UCI CKD records. The corresponding Brier score was 0.004 and logarithmic loss was 0.020. Together, these metrics document the discrimination and fitted-score performance observed in the evaluated cohort.

Table 1. Held-out source-cohort evaluation of the independently trained predictive models.

The laboratory processing and reporting implementation was additionally assessed using deterministic engineering records. These tests exercised unit conversion, quality filtering, derived-feature dependencies, model-artifact verification, missing-feature processing, model-specific result generation and isolation of patient-specific computational state. The verification suite recorded 64 passing tests.

9. Computational Efficiency and Comparative Technical Results

Computational performance was measured using common engineering fixtures passed through seven inference arrangements. Tests employed CPU execution, controlled numerical thread settings, five warm-

up batches, eight randomized execution rounds and three repetitions per round. The evaluated modes included independent workers, feature-minimal pipelines, full repeated-processing pipelines, integrated processing and preprocessing-sharing variants.

Inference arrangement	Median latency (ms)	p95 (ms)	Canonical conversions	Fitted transforms
Parallel workers	5.553	6.507	83	5
Sequential workers	21.628	35.240	83	5
Five feature-minimal pipelines	14.763	28.960	83	5
Five full pipelines	17.852	22.648	190	5
Integrated scalar system	12.271	30.058	38	4
Integrated system, separate transformations	14.688	16.164	38	5
Optimized independent pipeline	13.486	16.637	38	5

Table 2. Single-record computational comparison using the common engineering fixture.

For a complete five-output patient request, the integrated processing arrangement performed 38 canonical measurement conversions compared with 190 conversions across the repeated full-processing pipelines. The compatibility controller executed four fitted feature transformations for five independently trained predictors. These results demonstrate the reduction in repeated canonical processing and eligible numerical transformation work achieved by the described computational arrangement.

The conversion comparison measures the change in repeated input-normalization operations, while the fitted-transformation comparison independently measures the compatibility controller’s reuse decision. An optimized independent pipeline with shared ingestion also performs 38 canonical conversions for the same engineering fixture. The present architecture therefore combines the common conversion pathway with a separately verified fitted-transformation compatibility mechanism.

In fresh-process startup measurements, a single process loading the five fitted models recorded a median startup duration of 4,184.621 ms, compared with 5,657.357 ms for five separately initialized model processes. The recorded integrated-process peak resident set size was 308.9 MiB, and the sum of separately measured independent-worker resident-set peaks was 970.0 MiB.

Batch size	Parallel workers (ms)	Integrated scalar (ms)	Optimized independent (ms)
1	5.553	12.271	13.486
8	29.189	86.172	17.822
32	109.662	342.623	31.715

Table 3. Median end-to-end batch timing for selected execution arrangements.

The comparative results distinguish improvements in repeated feature processing from overall execution latency. The optimized independent batching arrangement also demonstrates the computational effect of vectorized model inference, while the integrated embodiment emphasizes compatible preprocessing reuse and independently preserved model execution.

In the measured single-record comparison, the integrated scalar pathway had median latency of 12.271 ms versus 17.852 ms for five repeated full-processing scalar pipelines and 14.688 ms for the same integrated pathway with transformation reuse disabled. Parallel independent workers achieved 5.553 ms. The integrated process recorded peak resident-set memory (RSS) of 308.9 MiB; the sum of independently measured worker RSS peaks was 970.0 MiB. These memory values describe their specified process-level measurement arrangements.

10. Technical Advantages of the Invention

The unified canonical laboratory representation enables accepted observations to be converted and processed through a common validation pathway for five independent disease assessments.

The dependency-controlled feature engine provides consistent calculation of reusable biochemical indicators across model-specific feature requirements.

The versioned registry and integrity checks establish verifiable relationships between registered feature schemas, fitted preprocessing arrangements and individual predictive model artifacts.

The compatibility-controlled reuse mechanism reduces repeat transformation work while preserving independently trained disease-specific predictors and calibrated output functions.

The system supports structured report generation with separately labeled endpoints, feature coverage, quality processing information and registered model provenance.

The implementation was exercised by 64 passing automated regression and integration tests covering typed observations, unit conversion, biochemical quality processing, essential and optional feature handling, transformation-key enforcement, request isolation, saved-model inference parity, artifact integrity checking and generation of the five-result report.

The implementation combines a single-request interface, isolated per-patient computational state and independently executable model components to support modular extension and maintenance.

11. Industrial Applicability

The invention is applicable to laboratory information middleware, health-data processing software, medical research platforms, model-serving infrastructure and integrated analytical reporting systems that employ multiple disease-specific classifiers. The disclosed modules may be executed on a general-purpose computing system, a local laboratory workstation or a suitable hosted processing platform receiving existing laboratory result data.

The arrangement is particularly useful where multiple independently trained predictive functions share biochemical inputs but require individually preserved feature schemas, fitted preprocessing parameters, endpoint definitions and calibrated model outputs.

12. Brief Description of the Drawings

Figure 1 illustrates the general arrangement of the integrated laboratory data processing system, showing canonical input processing, biochemical feature generation, independent predictive components and common report generation.

Figure 2 illustrates the flow of a patient record through quality verification, feature dependency evaluation, registry lookup, fitted transformation and disease-specific output assembly.

Figure 3 illustrates the computational compatibility controller, in which independently registered models access shared canonical features and reuse fitted numerical transformations under matching compatibility keys.

Figure 1. Integrated laboratory processing architecture

Input	Shared processing	Independent inference	Output
Existing blood and urine results	Validation → canonical units → accepted-feature map	Diabetes | Kidney | Liver	Five separately labeled assessments
Patient and collection metadata	Derived biochemical features → compatibility controller	Cardiovascular | Breast cancer	Integrated structured report

Figure 2. Request-specific processing sequence

Laboratory record → observation validation → canonical snapshot → quality processing → derived-feature calculation → verified model registry → ordered feature projection → compatible fitted transformation → independent prediction and calibration → combined report.

Figure 3. Transformation compatibility

Model registry → ordered input schema and fitted state → transformation compatibility key → request-local transformed-vector cache → independent disease-model predictor.

13. References

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14. Abstract

The present invention discloses an integrated computer-implemented system for processing existing blood and urine laboratory measurements and generating five independent disease-associated assessments. A canonical observation processor validates laboratory units, specimens, collection context and quality information and stores the accepted measurements in a verified snapshot. A biochemical feature engine

derives reusable laboratory indicators. A versioned registry identifies each independently trained predictive model, its ordered input schema, fitted preprocessing parameters, endpoint and calibration function. A compatibility-controlled execution mechanism reuses fitted numerical transformations within a patient request when the corresponding preprocessing-state keys match. Five disease-specific prediction components generate individually labeled results, which are assembled into a common electronic report. In a documented implementation, the architecture executes 38 canonical conversions and four fitted transformations in generating five independent assessments from a complete engineering input record.